

MAHMUD AFROZ

mahmudafroz@gmail.com • 479-304-9175 • [Linkedin](#) • [Google Scholar](#)

EDUCATION

Executive Master of Business Administration

August 2026

Angell Snyder School of Business
Ottawa University, KS
Specializing in Business Data Analytics

Master of Science in Geography

December 2023

Department of Geosciences
University of Arkansas, Fayetteville, Arkansas
Thesis title: Modeling the Peak Ground Acceleration and Assessing the Economic Loss and Casualty Due to Potential Earthquakes of Various Magnitudes in the New Madrid Seismic Zone
Funded by AmericaView, USGS

Bachelor of Science in Geography and Environment

July 2018

Department of Geography and Environment
University of Dhaka, Bangladesh
Major in Geography and environment. Minor in Geology, Soil, Water, and Botany

PROFESSIONAL EXPERIENCE

Geographic Information Systems Technician

March 2024 – Current date

Black Hills Energy, Fayetteville, AR

Lead GIS Technician overseeing the comprehensive management of GIS gas data for Colorado and part of Nebraska.

- Generate and perform gas utility data editing, map creation, and report generation.
- Facilitate communication between field operations and engineering teams.
- Maintain and update versioned GIS databases to ensure they reflect current information.
- Perform query and scripting operations for daily requests and maintenance.
- Manage utility projects, including analysis of engineering and CAD drawings.
- Provide mapping services across different departments within the company.
- Interpret as-built information accurately and incorporate it into GIS systems.
- Execute quality assurance and quality control processes to maintain data integrity and accuracy.

Library Geographic Information Systems Specialist

August 2022 – June 2024

Data Services, Mullins Library

University of Arkansas, Fayetteville, AR

Directed the standardization and development of GIS and ArcGIS training programs, significantly boosting data accuracy and user engagement across University Libraries.

- Collaborate with the Data Services Librarian and the Head of User Services/Physics Library to standardize GIS and ArcGIS training across University Libraries, reaching and positively impacting 100+ library staff members and users.
- Create and refine reusable lesson plans utilizing GitHub, Jupyter Notebooks, and Markdown.
- Formulate geospatial lesson plans covering diverse topics to train end-users on the following: Mastering Google Maps usage; demystifying the basics of mapping and GIS; Advancing ArcGIS skills; building engaging storyboards with ArcGIS; integrating data from other library databases and tools into maps; Using Python alongside ArcGIS.

Geosciences Computing Technician

August 2019 – August 2021

Department of Geosciences

University of Arkansas, Fayetteville, AR

- Led a cross-functional team of teaching assistants and faculty, ensuring optimal performance and functionality of the geospatial labs for 300+ students.
- Oversaw assistant ESRI ArcGIS online administrator duties, including creating and managing over 1300 ESRI accounts.

- Collaborated closely with the Center for Advanced Spatial Technologies (CAST) to manage student access to the CAST server space.
- Streamlined Windows installation through efficient workflows, orchestrating batch installations for multiple computers, resulting in around a 30% reduction in time and enhanced software deployment efficiency.
- Managed Windows licensing for the university, ensuring software compliance for 400+ computers and 100% compliance in software audits.
- Provided technical assistance to the Geosciences Department faculty, staff, and graduate students, resolving software compatibility issues and troubleshooting IT concerns.
- Ensured all users were equipped with the latest technology for classes by conducting regular tech updates.

Teaching Assistant

August 2019- May 2022

Department of Geosciences, University of Arkansas

- GEOS 5793 – Geospatial Unmanned Aircraft Systems
- GEOS 4583 – Enterprise and Multiuser GIS
- GEOS 3543 – Geospatial Applications and Information Science
- GEOS 3553 – Spatial Analysis Using ArcGIS
- GEOS 1111L – Physical Geology Laboratory

GIS Internships

ArkansasView

2020 & 2022

- Curated and published insightful remote sensing workflows on Applied Silvicultural Assessment (ASA) hazard map Wildfire in Arkansas, providing crucial data for wildfire management.
- Innovated by creating two new versions of GIS models compatible with recent versions of ArcGIS Desktop and ArcGIS Pro, enhancing user experience and efficiency.
- Leveraged GitHub for effective version control of workflows and documentation, ensuring the most up-to-date versions were always available.

CONFERENCE PRESENTATION

AmericaView Earth Sensors and Research Committee Meeting

November 2023

Title: Modeling the Peak Ground Acceleration & Assessing the Economic Loss & Casualty Due to Potential Earthquakes of Various Magnitudes in the New Madrid Seismic Zone

GSA 2022 North-Central/Southeastern Joint Section Meeting

April 2022

Cincinnati, Ohio

Title: Assessment of Economic Loss and Casualties Due to Earthquakes of Different Magnitudes at the New Madrid Fault Zone

HONORS AND AWARDS

Academic Scholarship from Malcolm K. Cleaveland Endowment in Geosciences, University of Arkansas: \$1000 in June 2020 and \$1250 in August 2021 for demonstration of innovative geospatial research on the impacts of earthquakes around the New Madrid fault zone.

TECHNICAL SKILLS

Geospatial	ArcGIS Pro, ArcGIS Desktop, ArcGIS Enterprise, ERDAS IMAGINE, pgAdmin, OpenJUMP, QGIS, Pix4D, Metashape
Asset management	LocusView, ABB Ability Asset Suite eSOMS (Electronic Shift Operations Management System)
Language	Python, C++, R, SQL, PostgreSQL, HTML & CSS
Desktop	GitHub, Zotero, MS Office, Adobe Suite
Data Visualization	Tableau and ArcGIS dashboards, Jupyter Notebook